BUSINESS ANALYTICS - INDIVIDUAL CASE STUDY

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Building India's Missing Poultry Price Panel: Forecasting NECC Egg Prices with a Walk-Forward Validated Baseline

Field	Detail
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Abstract

Indian poultry producers plan against a single daily declared egg rate because no public multi-year price series exists. This case study assembles that series by web scraping the National Egg Coordination Committee (NECC) daily zonal rate sheets, producing a panel of 195,519 zone-day price records covering 37 zones from January 2009 to July 2026. The panel is joined to a monthly feed-cost index and used for three tasks: measuring the price cycle, testing whether feed cost Granger-causes the egg price, and forecasting the zonal price 30 days ahead under walk-forward validation. The measured cycle shows that 11.4 per cent of all trading days fall inside a 30-day drawdown of 15 per cent or worse, with a seasonal swing of roughly plus or minus 16 per cent. A gradient boosted tree model reduces mean absolute error by 6.3 per cent against the persistence baseline a producer already has at no cost, reaching a mean absolute percentage error of 10.03 per cent. The causality hypothesis is not supported: no lag between one and twelve months reaches $p < 0.05$, and the negative result is reported as such. The main contribution is the reusable price panel and an honestly validated baseline against it.

Keywords: agri-commodity price forecasting, web scraping, walk-forward validation, Granger causality, gradient boosting, poultry supply chain.

1. Problem Statement and Objectives

1.1 The business problem

Indian poultry producers operate on thin margins in a market whose price movements are large and frequent. Decisions with multi-week consequences, such as flock sizing, feed procurement, contract pricing and distributor inventory, are taken against a spot rate that carries no information about the coming month. Three specific gaps follow from this.

1. No forward signal. The rate declared for today is the only number available to a producer. There is no published forecast of the zonal egg price at any horizon.
2. No public price history. Daily rates are published and then overwritten, so the series is not retained anywhere in public. Without an assembled panel, nobody can measure how often price falls occur, how deep they run, or whether they follow a seasonal pattern.
3. No margin buffer. Feed is approximately 70 per cent of poultry variable cost, with maize at 65 to 70 per cent and soymeal at 25 to 30 per cent of feed (ICRA, 2025). The diversion of maize to the ethanol programme raised maize used for ethanol to about 2.86 billion litres of output equivalent in ESY 2023-24, roughly 42 per cent of all ethanol, and India became a net maize importer in 2024 (Down To Earth, 2024). A price fall that was once a poor month is now a question of solvency.

1.2 Why the problem is worth solving

The volatility is measurable rather than anecdotal. Using the panel built for this study, 11.4 per cent of all trading days between 2009 and 2026 sit inside a 30-day drawdown of 15 per cent or worse, which is a price fall of that size on roughly one day in nine. The seasonal peak in November and December runs 15 to 17 per cent above the annual mean, and the trough in March and April runs 13 to 17 per cent below it. The persistent gap between the cheapest and the dearest zone is about 61 paise per egg. A producer carrying this exposure on a commodity where feed is 70 per cent of cost holds a risk that can be quantified but currently is not. Eggs are also a staple protein, so instability passes through to consumers and to institutional buyers.

A public price instrument is ordinary infrastructure rather than an unusual request. The USDA publishes a weekly Broiler Hatchery report of eggs set and chicks placed, and its Economic Research Service adjusts production forecasts using that data (USDA ERS, 2026). India has no equivalent, and the clearest evidence of the gap is that ICRA, a credit rating agency, cites a daily rate website as its data source in published poultry sector research (ICRA, 2025). Professional analysts read the same public web page a farmer reads.

1.3 Who benefits

- Layer farmers: flock sizing and feed procurement against a forward view rather than a spot rate.
- Egg traders and distributors: inventory and contract timing in a perishable, seasonally priced good.
- Feed mills: demand planning tied to bird numbers and producer margin.
- Institutional buyers such as schools, midday meal programmes and hospitality: quantified budget exposure to a volatile staple.
- Lenders and rating agencies: a reproducible sector stress series.
- Policymakers: an evidence base for feed, trade and producer distress measures.

1.4 Objectives

The study sets three objectives.

1. Build a multi-year Indian egg price panel by zone and day through web scraping, and use it to measure the price cycle: how often falls occur, how deep they are, and how the seasonal and regional structure is shaped.
2. Test whether feed cost leads the egg price, stated as two hypotheses. H1: feed cost Granger-causes the zonal egg price. H2: the cross-correlation of feed cost against egg price peaks at a positive lag, meaning feed cost moves first.
3. Forecast the zonal egg price 30 days ahead under walk-forward validation, and report skill against the persistence baseline a producer already has at no cost, so that the reported gain is the gain over current practice rather than over an arbitrary reference.

2. Data Collection and Dataset Description

2.1 Collection method

All primary data was collected by web scraping public web pages. No questionnaire or survey was used, and no ready-made dataset from Kaggle, UCI or a dataset repository was used as the primary source. The primary source is the NECC daily zonal rate sheet served at e2necc.com/home/eggprice. The page accepts an HTTP POST carrying a month and a year and returns a grid with one row per NECC zone and one column per day of that month. A scraper iterates over every month from January 2009 to July 2026, parses the grid, and writes one record per zone and date together with the source URL and the retrieval timestamp. A total of 212 source months were retrieved. A single auxiliary series is used for feed cost: monthly maize and soybean prices from the Federal Reserve Economic Data (FRED) service, combined into a feed-cost index. That is a published statistical series used as a secondary explanatory variable, not as the primary dataset.

2.2 Feasibility check carried out before scraping

The original design proposed reconstructing daily broiler and day-old-chick rate history from the Internet Archive. A coverage check against the Wayback CDX API found only about 33 captured days for the chick rate page and about 20 for the broiler page across 2019 to 2026, roughly sixty times too sparse for a daily panel: those committee-set rates are published fresh, overwritten daily, and were never preserved. The check confirmed the data gap that motivates the study, and it is why the study is built on the egg series, which was retained.

2.3 Dataset description

Property	**Value**
Primary source	NECC daily zonal rate sheet (e2necc.com), collected by scraping
Auxiliary source	FRED monthly maize and soybean series
Records	**195,519 zone-day observations**
Zones	37 NECC zones
Date range	1 January 2009 to 12 July 2026 (212 source months scraped)
Unit of price	Paise per egg, as declared
Feed coverage	99.3 per cent of rows
Engineered features	16 (lag, moving average, feed and calendar)
Held-out test observations	95,295 across five walk-forward folds

Table 1. Dataset summary.

Table 2 lists the attributes used for modelling. Attributes 1 to 5 are collected; the remainder are derived in the feature build step.

#	**Attribute**	**Type**	**Origin and meaning**
1	zone	Categorical	NECC zone label as published
2	date	Date	Trading date (ISO 8601)
3	egg_price	Numeric	Declared NECC zonal rate, paise per egg
4	maize_usd_mt	Numeric	FRED monthly maize price, USD per tonne
5	soy_usd_mt	Numeric	FRED monthly soybean price, USD per tonne
6	feed_cost_index	Numeric	0.70 x maize + 0.30 x soy, each mean-normalised
7	egg_ma7, egg_ma30	Numeric	7-day and 30-day moving averages, within zone
8	egg_lag1, egg_lag7, egg_lag30	Numeric	Price at t-1, t-7 and t-30
9	egg_yoy	Numeric	Price at t-365 (same day last year)
10	egg_ret30	Numeric	30-day return, price(t) / price(t-30) - 1
11	day_of_week, month	Categorical	Day of week (0-6) and calendar month (1-12)
12	sin_doy, cos_doy	Numeric	Cyclical encoding of day of year
13	target	Numeric	**egg_price at t+30, the forecast target**

Table 2. Attribute dictionary for the modelling panel.

2.4 Ethics and data handling

Only public web pages were accessed. No login or paywall was bypassed, robots.txt was respected, and requests were rate limited. The data consists of committee-declared market rates by zone and date. It contains no personal, individual or confidential information, so no anonymisation step was required. Every figure and number in this report is produced by code in the repository; none is entered by hand.

3. Data Preparation and Exploratory Analysis

3.1 Cleaning and preprocessing

1. Grid parsing and reshaping. Each monthly grid is converted from a zone-by-day matrix into long form, giving one row per zone and date.
2. Removal of non-trading cells. Blank and non-numeric cells, which correspond to holidays and days on which no rate was declared, are dropped rather than interpolated, so that no price is invented.
3. Range validation. Prices are retained only if they fall strictly between 0 and 2000 paise per egg. This removes parsing artefacts without trimming genuine extremes.
4. Sorting and per-zone grouping. Rows are sorted by zone and date, and every lag, moving average and return is computed within a zone, so that no feature crosses a zone boundary.
5. Feed join. The monthly feed series is joined to the daily egg series on year and month. This is a genuine granularity mismatch and is recorded as a limitation in Section 7.2.
6. Target construction and row removal. The target is the price 30 days ahead within the same zone. Rows without a target, or missing any required feature, are dropped before modelling. Because the year-on-year feature requires 365 days of history, the first year of each zone is not modelled.

3.2 Descriptive analysis and visualisations

Coverage was examined first, because a panel with intermittent reporting cannot support a daily forecast. The number of reporting zones rises from about 21 in 2009 to 34 by 2020 and remains continuous thereafter, with no gap in the daily series. Figure 1 then plots the national mean egg rate against the feed-cost index as two stacked panels rather than as one chart with two vertical scales, because a second scale can be chosen so that any two series appear related. The egg price shows a long upward trend with a strong repeating annual cycle. The feed index shows two distinct episodes, one around 2011 to 2013 and a larger one from 2021 to 2023. The two series do not move together in an obvious way, which is the first indication that the causal hypothesis tested in Section 4.4 may not hold.

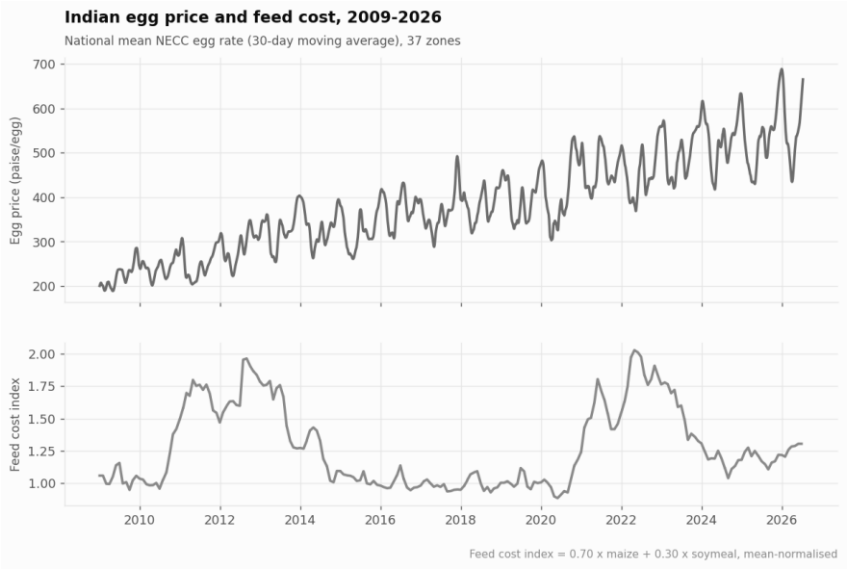


Figure 1. National mean egg rate and the feed-cost index, 2009 to 2026.

The price level is dominated by the long-run trend, which hides the seasonal cycle. Figure 2 shows the within-year profile as the median across all years with the interquartile band. The median price falls to about 310

paise in April and rises to about 420 paise in December, a difference of roughly 35 per cent between trough and peak. Measured against each year's own mean, the November to December peak is 15 to 17 per cent above the annual mean and the March to April trough is 13 to 17 per cent below it, and the pattern repeats in almost every one of the eighteen years. This places a constraint on the modelling: a model that only rediscovers that December is expensive adds nothing a producer does not already know.

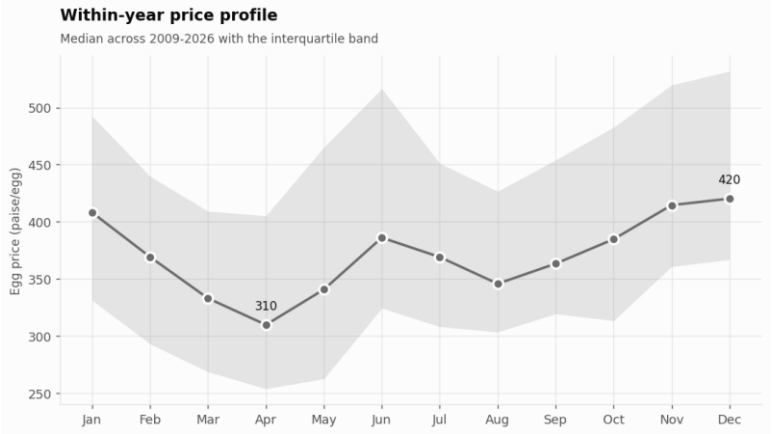


Figure 2. Within-year price profile: median across 2009 to 2026 with the interquartile band.

Figure 3 reports regional structure as each zone's mean gap to the national rate on the same day, rather than as a plain per-zone average. The correction matters because eight of the thirty-seven zones begin reporting only after 2015, and a plain average would rank them high simply for having a recent and higher-priced observation window. The corrected view places consumption centres such as the Kolkata belt, Mumbai and Pune at a premium, and production belts such as Ludhiana, Barwala and Hospet at a discount, which is the economically expected direction. The spread from Hospet at minus 27 paise to Midnapur at plus 34 paise is about 61 paise per egg, and it is persistent rather than transient.

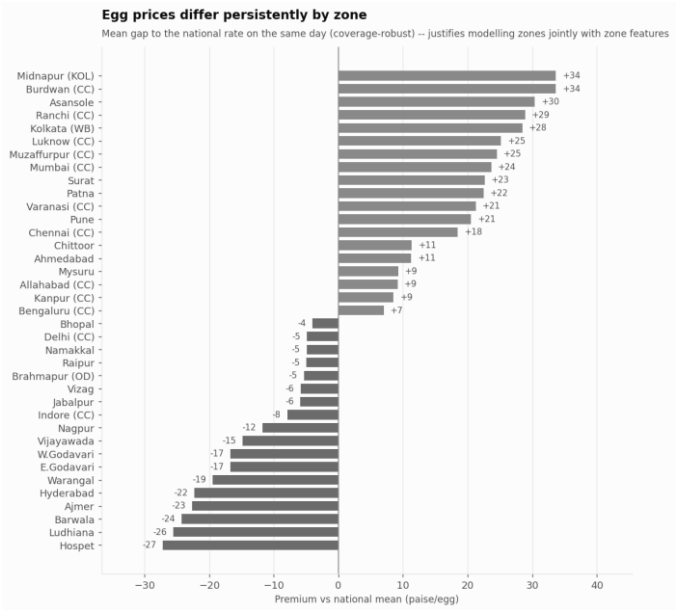


Figure 3. Mean premium of each zone against the national rate on the same day.

Figure 4 quantifies the phenomenon the study is about. The distribution of 30-day price changes on the national mean series is wide and has a substantial left tail: 11.4 per cent of all days sit at or below a 15 per cent fall over the preceding 30 days. This is the measured basis for the statement that a price fall of this size occurs on about one day in nine.

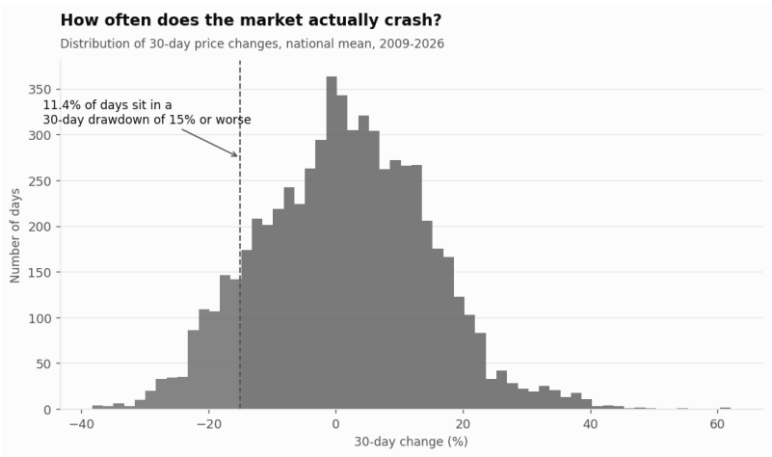


Figure 4. Distribution of 30-day price changes, national mean series, 2009 to 2026.

4. Analytics Method and Implementation

4.1 Choice of method and justification

The task is supervised regression on a panel time series: predict the egg price of a given zone 30 days ahead from information available on the day the forecast is made. Two model families from the Business Analytics syllabus are applied to the same feature panel.

- Ridge regression. A regularised linear model. It is included because it is interpretable, it remains stable when lag features are strongly correlated with one another, and it establishes whether a linear relationship alone is sufficient.
- Gradient boosted trees (XGBoost). A non-linear ensemble. It is included because the seasonal effect interacts with both the price level and the feed regime, and tree ensembles capture such interactions without the interaction terms having to be specified in advance.

Both models are trained on the same 14 numeric features listed in Table 2, so any difference in performance comes from the model rather than from the information supplied to it.

4.2 Baselines

A forecast is useful only if it improves on what a decision maker already has. Two naive baselines are therefore evaluated on exactly the same folds and metrics. Persistence predicts price(t+30) = price(t); this is the number a producer already has, at no cost, and a model that cannot beat it has no practical value. Seasonal naive predicts price(t+30) = price(t+30-365), that is, the same day last year.

4.3 Validation design

Validation is walk-forward, also called rolling origin, with five folds. The timeline is ordered by date, the first half is reserved as an initial training base, and the remaining half is divided into five consecutive test blocks. For each fold the model is trained only on dates strictly earlier than the start of its test block. Random k-fold cross-validation is not used at any point, because it places future observations in the training set and produces a score that could not be obtained in deployment. Table 3 shows the resulting test windows.

Fold	Test window	Test observations
1	18 March 2018 to 6 November 2019	17,361
2	7 November 2019 to 27 June 2021	16,769
3	28 June 2021 to 17 February 2023	20,400
4	18 February 2023 to 8 October 2024	20,366
5	9 October 2024 to 31 May 2026	20,399

Table 3. Walk-forward test windows. Training for each fold uses only earlier dates.

4.4 Causality testing

Hypotheses H1 and H2 concern the monthly relationship between feed cost and egg price, so they are tested on a national monthly mean series of 196 months rather than on the daily panel. Both series are first differenced to meet the stationarity requirement of the Granger test. The augmented Dickey-Fuller test on the

differenced series returns a p-value of 1.06 x 10^-8 for the egg price and 7.72 x 10^-18 for the feed index, so both are stationary after differencing. The Granger causality test is then run for lags of one to twelve months, and the cross-correlation of the differenced feed index against the differenced egg price is computed for leads of zero to twenty-four months.

4.5 Evaluation metrics and implementation

Three error metrics are reported: mean absolute error (MAE) in paise per egg, root mean squared error (RMSE) in the same unit, and mean absolute percentage error (MAPE). A skill score against the persistence baseline is also reported, defined as one minus the ratio of a model's MAE to the persistence MAE, so that a positive value is an improvement over current practice.

The pipeline is written in Python and runs from a clean state in four stages: scraping the NECC rate sheets, building the feature panel, fitting and evaluating the models with the causality tests, and generating the figures. It uses requests and BeautifulSoup for collection, pandas and NumPy for preparation, scikit-learn for Ridge, XGBoost for the boosted trees, statsmodels for the Dickey-Fuller and Granger tests, and Matplotlib for the figures. XGBoost is configured with 400 trees, maximum depth 6, learning rate 0.05 and subsample and column sample ratios of 0.8, under a fixed random seed; Ridge uses standardised features with a regularisation strength of 1.0. All metrics are written to a single machine-readable file, and the panel rebuilds to the same 195,519 rows.

5. Comparison with State-of-the-Art Methods

Four recent published studies on agricultural and egg price forecasting are compared below. Each was selected because it shares at least one element with this case study: the commodity, the country, the model family, or the use of a naive benchmark.

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Your Work
Meister and Yu (2025), Q Open, 5(2). Forecasting egg price inflation in Germany with machine learning	German packing-station egg prices (AMI); weekly; January 2008 to December 2022; 780 observations	LASSO feature selection, then ARIMAX and LSTM	RMSE, MAPE, Population Stability Index	ARIMAX consistently outperformed LSTM at the eight-week horizon. Predictors included imports, the cage ban, energy prices and bird flu	Same commodity, different market; a weekly national series rather than a daily zone panel. Both use an expanding-window split. Their exogenous set is richer than our feed-only set, and they report no persistence baseline
Manogna, Dharmaji and Sarang (2025), Scientific Reports. Enhancing agricultural commodity price forecasting with deep learning	AGMARKNET portal, India; 23 commodities; 165 markets; daily; January 2010 to June 2024	ARIMA, SVR, XGBoost, MLP, RNN, LSTM, GRU and Echo State Networks	RMSE, RNMSE, MAE, MAPE	GRU performed best, with MAPE of 14.59 per cent for onion and 10.58 per cent for tomato, against ARIMA above 92 per cent	Same country and daily frequency, but drawn from an existing portal rather than a scraped series, and validated on one 80/20 split instead of five rolling-origin folds. Our MAPE of 10.03 per cent is of similar magnitude but on a different commodity and split, so the two are not comparable
Mahajan, Muley and Fegade (2026), Discover Informatics, 1:15. Forecasting agricultural commodity prices using machine learning models	AGMARKNET, Maharashtra; 9 crops; monthly modal prices from daily mandi records; 2001 to October 2025	Random Forest, Extra Trees, Histogram Gradient Boosting and XGBoost against naive and seasonal naive baselines	MAE, RMSE, R-squared, MAPE	The one-month naive baseline gave the lowest MAPE for every crop, from 2.30 to 4.91 per cent. The ensembles beat the seasonal benchmark but not the naive one	The closest design to ours: same model family, same two baselines, same expanding-window scheme. Their models did not beat persistence, while ours beats it by 6.3 per cent MAE at 30 days. Their result supports treating persistence as the reference
Zhao, Chen,	Rice, wheat and	TCN-XGBoost	RMSE,	TCN-XGBoost	Shares the gradient-

Suraphee, Phoophiwfa and Busababodhin (2025), PLOS ONE. A hybrid TCN-XGBoost model for agricultural product market price forecasting	corn from World Bank, FAO, ITC and OECD sources; daily; January 2012 to December 2023; 4,380 days	hybrid against ARIMA, LSTM, Transformer-XGBoost and CNN-XGBoost	MAE, MAPE	reached RMSE 0.26 and MAPE 5.3 per cent, against LSTM at 8.1 per cent and ARIMA at 8.9 per cent	boosting component, but applies it to smoothed international index series with a fixed 70/15/15 split and no naive benchmark. The lower MAPE reflects a smoother target series, not transferability to a zone panel

Table 4. Comparison with recent published studies.

Two points follow from the comparison, and both are stated with care. First, the reported error figures are not comparable across these studies, because the commodities, the frequencies, the horizons, the markets and the validation designs all differ. A lower MAPE in one study is not evidence of a better method. Second, the one comparison that is methodologically close, Mahajan et al. (2026), reports that tree ensembles did not beat the naive benchmark on Maharashtra crop prices. That this study obtains a positive but small gain of 6.3 per cent over persistence is consistent with their finding rather than in conflict with it: short-run price persistence is strong in Indian agricultural price series, and any honest gain over it should be expected to be modest.

Relative to all four, the distinguishing feature of this case study is the dataset rather than the model. Each of those studies draws on a series that was already published as a series, whether by AMI, AGMARKNET or an international agency. This study constructs the series itself from daily pages that are overwritten.

6. Results, Business Insights and Recommendations

6.1 Forecasting results

Table 5 reports the mean of each metric across the five walk-forward folds, at a 30-day horizon, with the skill score measured against the persistence baseline.

Model	MAE (paise)	RMSE (paise)	MAPE (%)	Skill vs persistence
Naive (persistence)	50.75	63.46	11.10	reference
Naive (seasonal)	63.26	76.19	13.47	-24.7%
Ridge	49.76	60.44	10.52	+1.9%
XGBoost	**47.53**	**58.66**	**10.03**	**+6.3%**

Table 5. Mean performance across five rolling-origin folds, 30-day horizon.

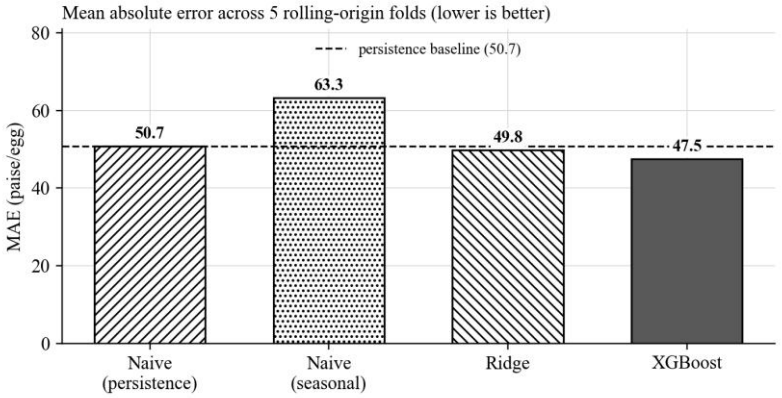


Figure 5. Mean absolute error by model against the persistence baseline.

Three results should be read together. XGBoost is the best model and beats the free baseline by 6.3 per cent in MAE, which is a real but modest improvement rather than a large one. Ridge improves on persistence by only 1.9 per cent, which indicates that a linear combination of the lag and calendar features adds very little. The seasonal naive baseline is 24.7 per cent worse than persistence, which shows that the year-on-year pattern alone is not a reliable predictor even though the seasonal cycle is clearly visible in Figure 2.

6.2 Stability across folds

The mean in Table 5 conceals a more important result. Figure 6 plots per-fold MAE, and the ranking of the models is not stable. XGBoost is the strongest model in folds 1 and 5, with MAE of 28.8 and 40.2 respectively, but in fold 2 it is the worst of the non-seasonal models at 68.9 against persistence at 53.1. Fold 2 covers November 2019 to June 2021, which contains the COVID-19 demand disruption. The fitted relationship did not survive that structural break. A single averaged number would have hidden this, and it is the main reason the model is presented as a baseline rather than as a decision tool.

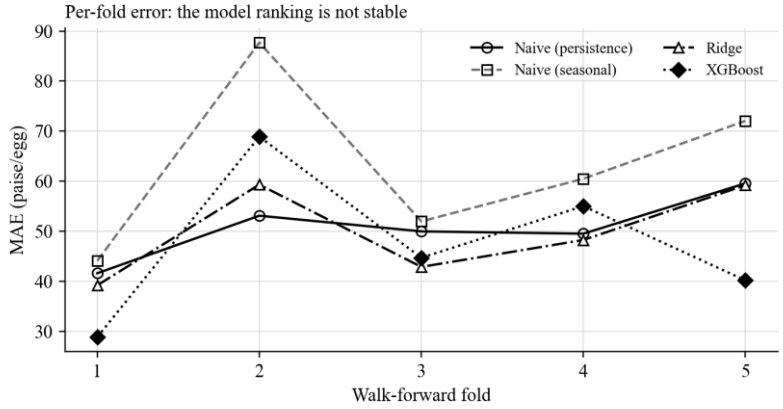


Figure 6. Per-fold mean absolute error. The model ranking changes between folds.

6.3 Error diagnostics

Figure 7 shows the residual distribution and the monthly error over time. Errors are wide and skewed to the left, with a mean bias of about minus 20 paise, meaning the model systematically under-predicts price peaks. For an early-warning application this is the safer direction of error, since it understates rather than overstates the price a producer can expect, but it is a bias and not noise, and it would make the forecast unsuitable for valuation or contract pricing without correction. The right panel shows that error is not uniform over time: monthly MAE rises sharply in 2020 and 2021, which is the same period identified in Section 6.2.

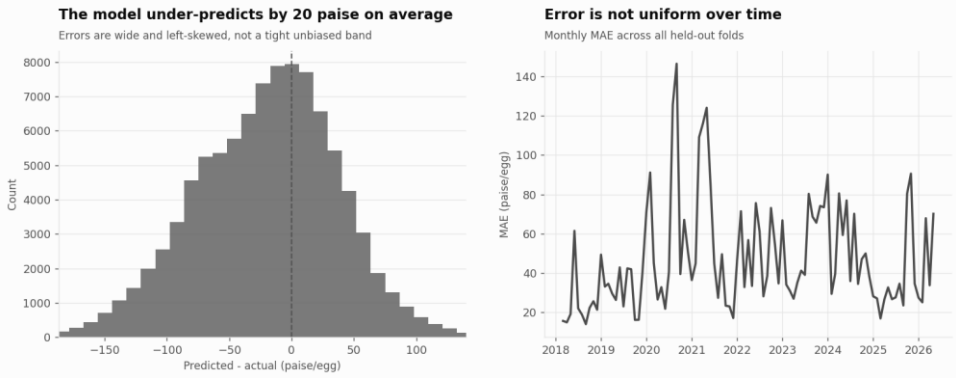


Figure 7. Residual distribution and monthly mean absolute error across all held-out folds.

6.4 Causality results: H1 is not supported

H1 is rejected at the conventional level. No lag from one to twelve months reaches $p < 0.05$. The smallest p-value occurs at a lag of seven months, at $p = 0.099$. On this data, feed cost does not Granger-cause the egg price at a five per cent significance level. H2 is weak. The cross-correlation peaks at seven months with $r = 0.22$. The sign is positive, which is consistent with feed cost leading rather than following, but the magnitude is too small to treat the lag as recovered.

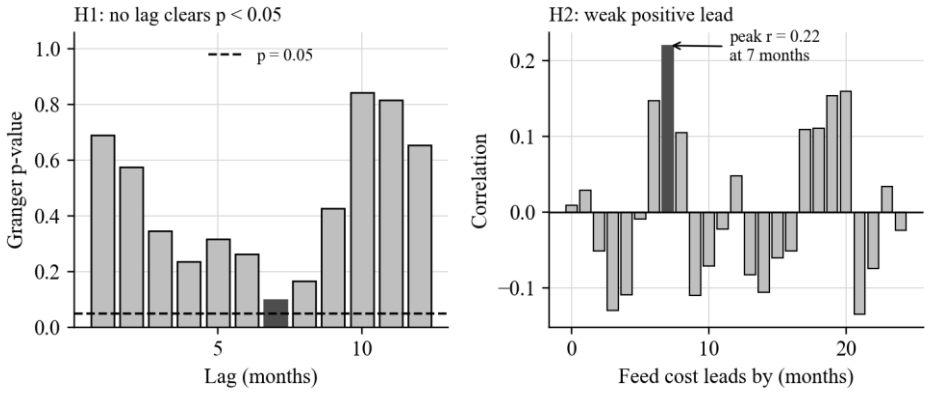


Figure 8. Granger p-values by lag (left) and cross-correlation of feed cost against egg price (right).

This is a negative result and it is reported as one. Three explanations are consistent with the data, and this study cannot separate them. Feed cost may genuinely not drive the egg price at this horizon; the monthly-to-daily granularity mismatch may destroy the signal; or international FRED prices may be a poor proxy for Indian mandi feed cost. The third is the most likely and is the first item for the next iteration.

6.5 What the model relies on

Gain-based feature importance for XGBoost shows where the model draws its signal. The previous day's price carries 0.264 and the calendar month carries 0.198, so recent level and season together account for close to half the explained gain. The three feed features, feed_cost_index at 0.125, maize at 0.051 and soy at 0.048, carry about 22 per cent in total. This is not negligible, and it is consistent with the weak Granger result: feed carries some predictive information without establishing causality.

6.6 Business insights

1. The price risk is quantified for the first time. A producer faces roughly a one in nine chance that any given day sits inside a 30-day fall of 15 per cent or more. This converts an informal expectation into a planning number.
2. Season is the largest reliable regularity, but it is already known to the trade. Prices are highest in November and December and lowest in March and April, with a swing of roughly plus or minus 16 per cent against the annual mean. A forecast therefore adds value only where it improves on the seasonal expectation, and the seasonal naive baseline shows that season alone is not enough.
3. Regional price levels are persistent and are worth planning around. The gap between the cheapest and the dearest zone is about 61 paise per egg and holds over years, with consumption centres at a premium and production belts at a discount.
4. A usable forward signal exists, but it is small. A 6.3 per cent reduction in mean absolute error over persistence, with the ranking unstable across folds, is a genuine improvement that is nonetheless too weak to drive a decision on its own.
5. Feed cost is not a usable leading indicator in the form tested. This matters commercially, because feed cost is the variable the trade watches most closely. The result argues for using domestic daily mandi feed prices rather than international monthly series before any conclusion is drawn about the relationship.

6.7 Recommendations

1. Use the forecast as a risk flag, not as a price. Given the minus 20 paise bias and the fold instability, the appropriate use is to signal an increased chance of a downward move over the next 30 days, which supports decisions on flock sizing and inventory, rather than to quote a number in a contract.
2. Plan procurement and inventory against the measured seasonal profile. The March to April trough and the November to December peak are stable enough across eighteen years to inform a purchasing and stocking calendar.
3. Replace the feed input before repeating the causality test. Daily Agmarknet mandi prices for maize and soymeal would remove both the granularity mismatch and the proxy problem, and would give H1 a fair test.
4. Publish and maintain the panel. The assembled series is reusable infrastructure. Continued daily collection extends it forward and increases its value to lenders, buyers and policymakers who currently have no series to refer to.

5. Treat structural breaks explicitly. Disease outbreaks, demand shocks and trade or ethanol policy changes are the events that broke the model in fold 2. Adding indicators for such events is the most direct route to a more stable forecast.

7. Conclusion and References

7.1 Conclusion

This case study set out to build a price instrument that did not exist publicly, and to report honestly what it supports. The instrument was built: a panel of 195,519 zone-day egg price records across 37 NECC zones from 2009 to 2026, assembled by web scraping daily rate pages that are otherwise overwritten, and joined to a feed-cost index. With it, the price cycle is measured rather than asserted: 11.4 per cent of trading days fall inside a 30-day drawdown of 15 per cent or worse, the seasonal swing is roughly plus or minus 16 per cent, and the persistent spread between the cheapest and the dearest zone is about 61 paise per egg.

On the modelling side the outcome is mixed and is reported as such. A gradient boosted tree model forecasts the zonal price 30 days ahead with a mean absolute percentage error of 10.03 per cent and reduces mean absolute error by 6.3 per cent against the persistence baseline, but the ranking is not stable across folds and the model under-predicts peaks by about 20 paise. The causal hypothesis that feed cost Granger-causes the egg price is not supported, with a best p-value of 0.099 at a seven-month lag, and the cross-correlation peak of r = 0.22 is too weak to recover the lag. The negative result is reported without restatement as a positive one.

The larger contribution is therefore the dataset rather than the model. The panel is reusable and reproducible from the code, and it supplies the measurement base that Indian poultry price analysis has lacked. The most useful next step is to replace the international monthly feed proxy with daily domestic mandi prices and repeat both tests on that input.

7.2 Limitations

- Feed data granularity and provenance. The feed index is built from monthly international FRED series joined to a daily Indian price by year and month. This is a mismatch on both axes and is the most likely explanation for the failed H1 test.
- Modest and unstable skill. A 6.3 per cent MAE gain that changes rank across folds is a weak basis for a production decision tool, and it is presented as a baseline.
- Administered prices. NECC zonal rates are declared by a committee rather than cleared purely by the market, which may attenuate the genuine supply and demand signal.
- Unequal zone coverage. Eight of the thirty-seven zones begin reporting only after 2015 and three stop early, so zone-level history is not uniformly deep. This is corrected where it matters, by same-day differencing in Figure 3.
- Systematic peak under-prediction. The minus 20 paise bias is acceptable for warning, but not for valuation or contracting.
- Structural breaks are not modelled. Disease outbreaks, demand shocks and policy changes are acknowledged as residual risk rather than represented as features.
- Scope. The findings concern layer and egg markets. They do not transfer directly to broiler production, whose cycle length and economics differ.

7.3 References

Published studies

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Data sources

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Industry and policy sources

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Code and reproducibility

[15] Case study repository, containing the scrapers, the feature build, the walk-forward model evaluation, the causality tests and the figure generation code: https://github.com/Goldmauler/poultry-cobweb-early-warning